

The Flavor and Charge Sectors: a spatio-temporal and Dirac–Kähler program for the emergent proton’s electroweak content

Tessera — cobordism subsystem (epic #410: #411–#418)

Abstract

The cobordism subsystem now carries the *color-confinement skeleton* of the proton: on the symmetric trivalent junction W_{ABC} a color-symmetric quark input transports to the net-color-zero singlet $\langle(1, \omega, \omega^2)\rangle$ with overlap 0.999, confinement ($\sigma \rightarrow 0$) is a Stokes conservation law at the junction, and the location of the singlet is forced by the geometry’s $\mathbb{Z}_3$ symmetry rather than imposed. What the construction does *not* carry is the electroweak content: electric charge, the proton/neutron (flavor) distinction, and the spinor/current structure. The earlier conclusion (#405) was that these need a *richer construction* — a flavor register parallel to color, plus spinor fields. This document lays out a different program, the one we are now building and testing: find the missing content by **splitting the behavior of the existing register** spatio-temporally and across form-degree, and by removing two arbitrary choices that currently break the geometry’s symmetry. Concretely: (i) source simplex orientation/sign from the standard simplicial formula, never a vertex-label sort; (ii) replace the asymmetric prism extrusion with a symmetric simplicial interior; (iii) read electric charge as a temporal-sector Gauss-law holonomy; (iv) realize the spinor/current content as the Dirac–Kähler factor of the Hodge operator; and (v) seek flavor in the spatio-temporal split or the Dirac–Kähler multiplicity, *not* a parallel register. We define each sector, its observable, and its falsifiable test; we fix a set of faithfulness invariants (G_1 – G_8) that no subtask may regress; and we give the dependency structure that says what can be built in parallel. This is a *plan*, honest about what is established versus hypothesis; the measurements live in the subtasks.

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1 Where we are, and what is missing

Established. Matter is not imposed in this subsystem. Color states are pinned on the boundary ∂W of a Lorentzian Regge cobordism W , the interior edge lengths relax to a stationary point of the dual Regge action, and the emergent result is *read off* the relaxed geometry. Color lives in the homology of a holed register surface; the proton is the trivalent junction W_{ABC} — one connected 2-frequency geodesic icosahedron (S^2 , 42 vertices) minus twelve vertex-disjoint hole triangles in four windows of three (A, B, C inputs and R the emergent result), extruded $\times I$. Three facts are in hand:

- **The emergent color singlet.** With the windows placed as one tetrahedral A_4 orbit, the input $\rightarrow$ result transport M intertwines the color $\mathbb{Z}_3$, $MP_{\text{in}} = P_{\text{out}}M$, and the color-symmetric (ω -representation) input transports to the singlet line with overlap 0.999.
- **Confinement as conservation.** The global Stokes relation $\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$ holds exactly ($\sim 10^{-14}$ on the uniform metric); a net color charge has no realizable harmonic representative.
- **The right dimensionless ballpark.** Read emergently, $\sigma \rightarrow 0$, and $r \cdot m \approx 6.9$ against the physical proton’s 4.0.

The boundary (#405). Every electroweak quantum number needs structure the *color-only* geometry does not carry. The three windows are identical color registers, so nothing distinguishes the proton (uud, +1) from the neutron (udd, 0): that is *flavor*. Electric charge is flavor-dependent (u: $+\frac{2}{3}$, d: $-\frac{1}{3}$). Spin needs spinor content; the construction is a geometry of scalar/ complex cochains. The natural “net orientation as charge” read reduces to the (zero) color flux or a particle/antiparticle sign — not the flavor-dependent electric charge.

The fork. The earlier write-ups proposed the minimal extension as a *richer construction*: a flavor register parallel to color, and spinor fields. We take a different road, for two reasons. First, a second hole register is geometrically expensive and competes with color for the surface’s vertex capacity, and if it shares the color holes it is no longer independent — it double-counts. Second, and more importantly, the missing content has a natural home in structure the construction *already has* but does not yet use: the Lorentzian (temporal/spatial) split of its cochains, and the higher-degree pieces of the cochain complex. The program of this document is to use that structure.

2 The reframe

Principle 1 (Split the register, do not multiply it). *The proton’s electroweak content is sought by splitting the behavior of the existing register — across causal type (temporal vs. spatial) and across form-degree (the inhomogeneous cochain) — rather than by adding parallel hole registers. Independence between sectors is to be supplied by orthogonality in the spectral/causal decomposition, not by disjoint topology.*

Three observations make this concrete.

Charge is a temporal-sector object. The electromagnetic field strength is a 2-cochain $F \in \Omega^2$. On a Lorentzian complex it splits by the causal type of the plaquette: the *electric* field is the part with a timelike leg, the *magnetic* field is the purely spacelike part. Electric charge is then Gauss’s law on the temporal sector, $d\star F = \star j$ with $\oint \star F = Q$ — a genuine gauged- $U(1)$ holonomy, quantized and metric-robust, not a hand-chosen weight. The quark charges are multiples of $\frac{1}{3}$, and the $\frac{1}{3}$ -quanta are tied to *triality* (the color $\mathbb{Z}_3$ center the construction already carries) through Gell-Mann–Nishijima $Q = I_3 + \frac{1}{2}Y$.

Spin/current is the Dirac–Kähler factor of the operator. The inhomogeneous cochain $\Omega^\bullet = \bigoplus_k \Omega^k$ is the Dirac–Kähler field; the operator $(d + \delta)$, whose square is the Hodge Laplacian $L = (d + \delta)^2$, is the discrete Dirac operator, with gamma matrices realized as the Clifford action of edges (1-cochains). In four dimensions $\dim \Omega^\bullet = 16 = 4 \times 4$ and the operator algebra is the Dirac algebra $\mathbb{Cl}(1, 3) \otimes \mathbb{C}$: a genuine 4×4 gamma structure, *factored out of the Hodge operator the construction already builds*. Its conserved current’s time component j^0 is the charge density. This is also the principled distinguished frame the deferred operator read-out lacks (§5).

The geometry must be honest. Two arbitrary choices currently break the symmetry the physics needs. The simplex orientation/triangulation is in places decided by a vertex-label `std::sort`, not the standard simplicial orientation; and the $\times I$ prism extrusion is asymmetric — it is the documented source of the $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\| \approx 4.3 \times 10^{-2}$ intertwining residual that pins the singlet at 0.999 instead of 1. Both must be removed before the spectral/causal split can be trusted.

The remainder of this document develops these into a buildable program. Each section names the subtask issue that carries it and the falsifiable test that keeps it faithful.

3 Geometric foundations

The two foundations remove arbitrariness; everything downstream inherits the symmetry they restore.

3.1 Standard simplex orientation, not label sort (#412)

The standard orientation of a k -simplex is an equivalence class of vertex orderings under *even* permutations; the sign of a given ordering is the parity of the permutation relative to a *consistent induced (global) orientation*. Choosing the sign by sorting *vertex labels* makes the result depend on the arbitrary numbering of the complex, which is not a symmetry of the geometry. The icosahedral automorphism g that realizes the color $\mathbb{Z}_3$ permutes vertices but not labels; a label-sort sign is therefore not g -equivariant, and the symmetry leaks into the read-out.

Principle 2 (Orientation is geometric). *Every simplex orientation/sign — in the boundary operator, the period read-out covector, and any triangulation diagonal — is sourced from the standard simplicial orientation formula tied to the manifold’s induced orientation (`SimplexOrientation::orientationOf`). Sorting survives only as a sign-neutral canonical storage key, never as a sign.*

Faithfulness. The test of correctness is *relabeling-invariance*: applying a random permutation to the base-surface vertex ids and rebuilding W_{ABC} must leave the color charge σ , the singlet overlap, and the intertwining residual unchanged up to one consistent global sign shared by all windows. This is the invariant G_6 of §9; #412 owns it.

3.2 A symmetric simplicial interior (#413)

The cobordism interior is currently the Freudenthal “staircase”: each base cell is cut into simplices along a diagonal selected by sorting vertex ids. That cut is label-dependent and not g -invariant, and it — not the metric — is the cause of the 4.3×10^{-2} residual: replacing the uniform seed by a g -invariant non-degenerate metric leaves the residual essentially unchanged. The fix is a *symmetric continuation of the primal 1-skeleton*: a triangle whose base sits at $t = 0$ has its apex (tip) at $t = 1$, and that tip is shared with an inverted triangle based at $t = 2$ — the alternating up/down tetrahedral (tetrahedral–octahedral) stacking. It is label-independent and yields genuine vertical worldline (temporal) edges.

Remark 1 (Connectivity is CDT, the metric stays Regge). The alternating stacking is the Ambjørn–Loll Causal Dynamical Triangulations *connectivity*; the codebase already carries that orientation convention. But we do *not* import CDT’s preferred time foliation: the edge lengths stay free, the causal type emergent, the action stationary. Connectivity and the metric are independent choices; we take the symmetric connectivity and keep the Regge metric.

This foundation does double duty. It removes the residual (driving the singlet overlap to 1), and its worldline edges are the clean temporal edges the charge sector needs. The one care point is the octahedral cells of the stacking: they must be tetrahedralized without reintroducing a label-dependent diagonal (use the orientation anchor of §3.1, or symmetrize over the octahedral diagonals, or document the residual symmetry honestly).

Faithfulness. The residual $\|MP_{\text{in}} - P_{\text{out}}M\| / \|M\|$ must drop well below the 4.3×10^{-2} prism value and the singlet overlap rise above the 0.999 floor, while $b_1 = 11$, the dual complex stays a valid manifold, and charge conservation stays exact. #413 owns “singlet $\rightarrow 1$ ”.

4 The charge sector

4.1 E and B from the temporal/spatial split (#417)

Let $A \in \Omega^1$ be a $U(1)$ connection and $F = dA \in \Omega^2$ its curvature. On the Lorentzian complex each plaquette (a 2-cell, dually a hinge) is classified by the causal type of its edges:

$$F = \underbrace{F^{(E)}}_{\text{plaquettes with a timelike leg}} \oplus \underbrace{F^{(B)}}_{\text{purely spacelike plaquettes}}, \quad (1)$$

the discrete analogue of $E_i = F_{0i}$, $B_{ij} = F_{ij}$. The split is read off the existing geometry by causal type — no new register, no new holes. The reader is a pure observable: it must leave every color observable of §1 bit-for-bit unchanged.

Faithfulness. On a purely spacelike (static) seed $\|E\| = 0$ and B carries the structure; introducing timelike flux makes $\|E\| > 0$; and $F = dA$ is gauge-invariant ($A \rightarrow A + d\chi$ leaves E, B unchanged). A wrong causal partition leaks B into E and the static-seed test catches it.

4.2 Charge as a Gauss-law holonomy (#411)

Electric charge is the discrete Gauss law: the flux of E through a closed surface enclosing the quark windows,

$$Q = \oint_{\partial V} \star F = \int_V \star j, \quad (2)$$

the time component of the conserved current. The point is what kind of object this is. The previously-considered “flavor-weighted covector” $(\frac{2}{3}, \frac{2}{3}, -\frac{1}{3})$ is not A_4 -equivariant and is not the topologically protected functional: it would degrade under a non-symmetric seed the way charge conservation degrades from 10^{-14} to $\sim 10^{-2}$ under jitter. The Gauss-law holonomy (2), by contrast, is a genuine gauged- $U(1)$ invariant — quantized and metric-robust.

Triality and fractional charge. Quark electric charges are multiples of $\frac{1}{3}$, and in the Standard Model the fractional part is tied to color through the center $\mathbb{Z}_3$ (triality) via $Q = I_3 + \frac{1}{2}Y$, with $\frac{1}{2}Y$ carrying baryon number/triality. The construction already carries triality (the color $\mathbb{Z}_3$ is P_{out}). This sector investigates whether the $\frac{1}{3}$ -quantum of the Gauss-law charge *is* that triality — i.e. whether fractional electric charge is forced by the same $\mathbb{Z}_3$ that forces confinement.

Faithfulness. Over an ensemble of metric jitters, $\text{std}(Q)$ stays tiny while the covector baseline drifts by orders of magnitude (the ratio is the figure of merit); Q lands on the third-integer lattice; and on the color singlet Q equals the expected neutral total.

5 The spin/current sector: Dirac–Kähler

5.1 The operator and its gamma structure (#415)

The inhomogeneous cochain $\Omega^\bullet = \bigoplus_k \Omega^k$ is the discrete Dirac–Kähler field. With the coboundary d and its adjoint (codifferential) δ , the Dirac–Kähler operator is

$$D = d + \delta, \quad D^2 = (d + \delta)^2 = d\delta + \delta d = L, \quad (3)$$

the Hodge Laplacian. Thus D is a *square root* of the operator the construction already assembles. Clifford multiplication by a 1-cochain e acts on $\Omega^\bullet$ by $e \vee \alpha = e \wedge \alpha + \iota_e \alpha$ (an edge raises and lowers form degree), and these satisfy the Clifford relations

$$\{\gamma_a, \gamma_b\} = 2\eta_{ab}. \quad (4)$$

In four dimensions $\Omega^\bullet \cong \mathbb{C}l(1, 3) \otimes \mathbb{C} \cong M_4(\mathbb{C})$, the 4×4 Dirac algebra: the gamma matrices are *factored out of the operator*, exactly the intuition that the spin matrix is a factor of the (Choi/operator) object of the right dimension.

The distinguished frame. The operator read-out $U = \text{unvec}(\ker L_1(W - \partial W))$ is presently deferred because the kernel basis is fixed only up to an $O(d^2 - 1)$ rotation — there is no canonical frame. The Dirac–Kähler structure supplies one: the Clifford generators are a distinguished basis with distinct eigenstructure, exactly as P_{out} ’s distinct eigenvalues $\{1, \omega, \omega^2\}$ gave color its canonical frame. Pinning the operator frame to the gamma generators is the same de-framing trick, transplanted from the state sector to the operator sector.

5.2 The current as charge density, and the multiplicity

The Dirac–Kähler current j has time component j^0 equal to the charge density; this must agree with the Gauss-law charge of §4.2, tying the operator sector to the temporal sector. The construction has a known feature: the Dirac–Kähler field in d dimensions carries $2^{d/2}$ Dirac spinors — a four-fold multiplicity in 4D (the “doubling”). On the lattice this is usually a nuisance; here it is a candidate *flavor/taste* index, a built-in internal multiplicity that may furnish flavor without a second register (§6). We record it honestly as a hypothesis, not a result.

Remark 2 (Spin is an operator property; color/flavor are state properties). This sector placement is not arbitrary. Spin is how a state transforms under rotation — an *operator*. Color (and, below, flavor) are properties of the boundary state’s periods. The qudit form of the singlet obstruction already hints at this: the qubit ($d=2$) alternating invariant ε_{ij} is the spin singlet, bipartite, living in the $d=2$ operator sector, while the qutrit color singlet ε_{ijk} is irreducibly tripartite in the state sector.

6 The flavor sector

Proton (uud) and neutron (udd) differ only by flavor; the color-only geometry cannot tell them apart. By Principle 1 we seek flavor as a *split of the existing register*, not a parallel hole register. Two candidate mechanisms are on the table:

- (i) the spatio-temporal split — does the temporal versus spatial behavior of the register furnish a binary, u/d-like label?
- (ii) the Dirac–Kähler multiplicity (§5) — does its four-fold taste index carry flavor?

The success criterion is concrete and shared with the charge sector: a geometric handle that yields total electric charge +1 for uud and 0 for udd.

Remark 3 (Why “flavor as a second nonzero spectral band” is rejected). A tempting reading of “flavor as a second harmonic” is the first *nonzero* eigenvalue band of L_1 . This is unsound. A period/holonomy is defined only on *closed* forms (the kernel); a $\lambda \neq 0$ eigenform has $d\psi \neq 0$, so its “period” is representative-dependent and not Stokes-conserved — and the charge sum of §4 type-checks only on the kernel. A generic nonzero band is also multiplicity one, so it cannot host an isospin doublet. Flavor must be either a second *kernel* structure or, as proposed here, the spatio-temporal / Dirac–Kähler split — not a free-floating band.

Faithfulness. The discriminator $Q(\text{uud}) = +1$, $Q(\text{udd}) = 0$ is the forward test. The decisive *structural* guard is that b_1 stays 11: if a new independent cycle appears, a register has been smuggled in, contradicting Principle 1. Proton and neutron must differ only in the flavor label, not in color content ($\sigma \rightarrow 0$ for both).

7 The binding structure: a bipartite revisit (#416)

The proton was placed on a trivalent junction because the color singlet ε_{ijk} is absent from $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and first appears in $\mathbf{3}^{\otimes 3}$. That theorem rules out a *generic* bipartite merge: the relaxed two-quark result carries $\mathbf{6} \oplus \bar{\mathbf{3}}$ and the symmetric $\mathbf{6}$ contaminates. It does *not* rule out two bipartite steps with a projection: if the first merge projects onto the antisymmetric diquark $\bar{\mathbf{3}} = \varepsilon qq$, the second ($\bar{\mathbf{3}} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8}$) contains the singlet.

Two questions follow, both testable with machinery that already exists — the prism extrusion accepts a cumulative *twist* permutation, i.e. an orientation-reversing tube that is a geometric antisymmetrizer.

- Does a twisted bipartite tube project the emergent diquark onto $\bar{\mathbf{3}}$ ($|\sigma| \approx 1$, the antisymmetric channel) rather than the generic average ($|\sigma| \approx 0.67$)?
- The deeper obstruction is that the colored diquark intermediate is non-carriable (confinement): only $\sigma = 0$ combinations carry as free boundary states. A *connected* two-vertex “path” geometry, where the diquark is internal and never a free boundary, evades this. Does such a path reach the singlet, and is it genuinely distinct from the symmetric star junction?

This sector is included because it bears directly on whether the proton’s binding (and thus its baryon number) is irreducibly a star or admits a path realization — a structural question underneath flavor and charge.

Faithfulness. The twisted diquark’s $|\sigma|$ must clearly separate from the generic value; the path geometry’s singlet overlap is pinned to a recorded verdict; the intermediate’s carriability residual is reported so a “separated two-step” claim cannot quietly rely on a non-carriable free diquark.

8 The dimensional question (#418)

The full structure of this program wants more than the current geometry supplies. The electromagnetic field strength has its full $\vec{E}, \vec{B}$ content, and the Dirac algebra its full 4×4 gammas, only in $3+1$ dimensions — a genuine S^3 spatial slice extruded in time. The present construction is $2+1$ dimensional ($S^2 \times I$), a reduced sector: in $2+1$ D the field strength has three components (B is a scalar) and the Clifford algebra is $Cl(2,1)$ (2×2 gammas). This is a spike, not a build: it scopes what the Hodge/Regge/relaxation machinery needs to generalize to a triangulated S^3 , the cost, and — crucially — which observables genuinely *require* $3+1$ D versus which are faithful already in $2+1$ D (the Gauss-law charge and the binary flavor label can be explored first in $2+1$ D; full $\vec{E}, \vec{B}$ and 4×4 Dirac–Kähler cannot). The deliverable is a go/no-go recommendation and a minimal S^3 prototype with a smoke test.

9 Faithfulness and anti-drift

This is a multi-step program over one shared geometry; later subtasks must not silently regress earlier results. We fix the *golden* invariants — the proton facts already established — as a shared regression guard, and require every subtask’s tests to keep them.

	invariant	assertion (the “do not drift” guard)
G_1	proton present	singlet overlap ≥ 0.999 (prism), $\rightarrow 1$ after #413
G_2	confinement	color charge $\sigma \rightarrow 0$ ($ \sigma \lesssim 0.06$) for the symmetric input
G_3	charge conservation	$\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$ exact ($\sim 10^{-14}$, uniform metric)
G_4	topology	$b_1 = 11$ and the dual complex is a valid manifold (no weld)
G_5	color $\mathbb{Z}_3$ intact	P_{out} eigenvalues $\{1, \omega, \omega^2\}$
G_6	relabeling-invariance	permuting vertex ids leaves σ /overlap/residual invariant (owned by #412)
G_7	determinism	fixed seed $\Rightarrow$ reproducible result; no wall-clock/random seeding
G_8	emergent-first	observables read off the relaxed geometry, never hand-placed

Each subtask carries a *Faithfulness & anti-drift tests* section that pins its own forward test (expected values, fixed seed) and names the subset of G_1 – G_8 it must keep. They share one module, `tests/cobordism/test_epic410_invariants.py`; every pull request in this track must keep it green. The discipline is sector-specific: an observable-only sector (§4.1) must leave G_1 – G_5 bit-for-bit unchanged; a geometry sector (§3.2) must preserve G_3/G_4 ; the flavor sector (§6) must keep b_1 unchanged so flavor is not smuggled in as holes.

10 Dependency structure and sequencing

The subtasks form a small dependency graph; “ $A \rightarrow B$ ” means B needs A .

```
#412 orientation --> #413 interior ---> #417 E/B --> #411 charge ---
                                     +--> #415 Dirac-Kahler ----> #414 flavor
#416 bipartite twist      (soft: #412 clean signs)
#418 S^3 spike           (investigation; informs #417/#411/#415)
```

The parallelization waves:

- **Wave 0** (start in parallel): #412 (orientation, the root), #416 (bipartite, independent), #418 (S^3 spike, early so it can scope the rest).
- **Wave 1**: #413 (symmetric interior), after #412.
- **Wave 2** (parallel): #417 (E/B) and #415 (Dirac–Kähler), after #413.
- **Wave 3**: #411 (Gauss-law charge), after #417.
- **Wave 4**: #414 (flavor, the proton/neutron discriminator), after #411 and #415.

The edges are forced: #413 needs #412 so the symmetric tetrahedralization does not re-import a label-sort diagonal; #417 needs #413’s worldline edges for a clean causal split; #411 needs #417’s E ; #414’s success metric is the #411 charge and its second mechanism is the #415 multiplicity.

11 Open questions and honest risks

This program is a plan; several of its load-bearing claims are hypotheses that the subtasks must confirm or kill.

- **Spin-statistics / total antisymmetrization (the deepest gap).** A proton is the totally-antisymmetric $SU(6)$ qqq state: color (already antisymmetric, the ε_{ijk} line) *forces* flavor $\times$ spin $\times$ spatial to be symmetric and entangled (the **56**-plet). $|p\uparrow\rangle$ does not factorize into flavor $\otimes$ spin. Three sectors treated independently — color (state), flavor (state), spin (operator) — supply no mechanism for that cross-sector entanglement; placing spin in the operator sector and flavor in the state sector arguably forbids it. This is the program’s largest unresolved question and must be held in view throughout.
- **Isospin is $SU(2)$, not a $\mathbb{Z}_2$ label.** A faithful flavor sector needs the isospin $SU(2)$ to *act* (the ladder $p \leftrightarrow n$), as the color $\mathbb{Z}_3$ acts via the P_{out} intertwiner. A static binary u/d label is weaker; whether the spatio-temporal or Dirac–Kähler split furnishes a genuine doublet with a ladder is open.
- **The Dirac–Kähler doubling.** The four-fold multiplicity is a feature only if it can be *interpreted*; it may instead be a discretization artifact. Its status as flavor/taste is a hypothesis.
- **Spinor double cover.** Genuine spin- $\frac{1}{2}$ requires the central $\mathbb{Z}_2$ of $\text{Spin} \rightarrow SO$, distinct from the determinant/orientation $\mathbb{Z}_2$; the Dirac–Kähler construction must be checked to realize the former (a 2π rotation $\mapsto -1$), not merely the latter.

- **Metric dependence.** The temporal/spatial split and the nonzero spectral content are metric-dependent and move under relaxation, unlike the topological color kernel. This is physically apt (electric/flavor structure is not exactly conserved) but demands care that observables are read at a well-defined point and are not drifting artifacts.

12 Scope and relationship to the existing program

This is its own exploration track (epic #410), independent of the proton epic's (#380) sequencing and deferrals. It does not modify the color result; it builds outward from it. The geometric foundations (§3) also improve the color result directly (the symmetric interior drives the singlet overlap to 1), so they are worth doing regardless of how far the electroweak sectors reach. The honest framing is the same as the color program's: we read what the geometry produces, we do not impose the answer, and we let the falsifiable tests of §9 arbitrate.